## Larkhall NDE Services Inc.

**Inspection Report**

Site: Hamilton Plant

| No. | Status | Date |
|---|---|---|
| LNDE-RPT-20160118-004-DS-6-SG-101 | Issued | 2016-02-12 |

# UT Thickness Data Sheet - 6-SG-101

### Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for the sour gas inlet line 6-SG-101 located within the ATU-100. The scope of work is limited to the thickness data collection required for the full-unit survey as detailed in LNDE-RPT-20160118-004-R0.

Marcus Chen, acting as the lead UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., is responsible for the execution of the measurements and the recording of the data presented herein.

### Technical Basis

Thickness measurements were performed in accordance with the site NDE procedure QSCP-NDT-001 Rev 2, which incorporates requirements for insulation removal and inspection-port access as part of the CUI program.

The equipment calibration and measurement techniques were conducted in accordance with the requirements of ASTM E797.

### Component Design Data

The subject line 6-SG-101 is constructed from A106 Gr.B with a nominal outer diameter of 168.3 mm and a nominal wall thickness of 7.11 mm. The design pressure is 1965 kPa(g) and the design temperature is 66 °C.

The minimum allowable thickness (t-min) for this line is 3.56 mm. Any reading recorded below this retirement thickness shall be raised as an NCR.

### Inspection Interval and Classification

Per the survey requirements in LNDE-RPT-20160118-004-R0, this line is subject to the thickness and external visual inspection intervals defined for its service class. As a sour gas line, it is categorized under Class 1, requiring a 5 year interval.

The remaining life and the date for the next thickness reading are determined by the larger of the long-term or short-term corrosion rate, with the next measurement due at no more than half the remaining life and within the class maximum.

### Thickness readings, measured 2016-02-12

**Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-SG-101-01 | 7.11 | 3.56 | 2011-02-18 | 6.39 | 6.17 | 0.048 | 0.044 | >50 | 2021-02-12 | |
| 6-SG-101-02 | 7.11 | 3.56 | 2011-02-18 | 6.14 | 5.76 | 0.065 | 0.076 | 28.8 | 2021-02-12 | |
| 6-SG-101-03 | 7.11 | 3.56 | 2011-02-18 | 6.17 | 5.87 | 0.061 | 0.060 | 38.1 | 2021-02-12 | |
| 6-SG-101-04 | 7.11 | 3.56 | 2011-02-18 | 6.05 | 5.58 | 0.076 | 0.094 | 21.4 | 2021-02-12 | |
| 6-SG-101-05 | 7.11 | 3.56 | 2011-02-18 | 5.87 | 5.38 | 0.088 | 0.098 | 18.5 | 2021-02-12 | |
| 6-SG-101-06 | 7.11 | 3.56 | 2011-02-18 | 6.27 | 5.88 | 0.060 | 0.078 | 29.6 | 2021-02-12 | |
| 6-SG-101-08 | 7.11 | 3.56 | 2011-02-18 | 6.33 | 6.00 | 0.066 | 0.066 | 36.8 | 2021-02-12 | |

**Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-SG-102-01 | 7.11 | 3.56 | 2011-02-18 | 5.75 | 5.42 | 0.086 | 0.066 | 21.6 | 2021-02-12 | |
| 6-SG-102-02 | 7.11 | 3.56 | 2011-02-18 | 5.80 | 5.43 | 0.084 | 0.074 | 22.2 | 2021-02-12 | |
| 6-SG-102-03 | 7.11 | 3.56 | 2011-02-18 | 5.80 | 5.22 | 0.091 | 0.116 | 14.3 | 2021-02-12 | |
| 6-SG-102-04 | 7.11 | 3.56 | 2011-02-18 | 5.73 | 5.18 | 0.091 | 0.110 | 14.7 | 2021-02-12 | |
| 6-SG-102-05 | 7.11 | 3.56 | 2011-02-18 | 6.46 | 6.19 | 0.047 | 0.054 | 48.5 | 2021-02-12 | |
| 6-SG-102-06 | 7.11 | 3.56 | 2011-02-18 | 6.41 | 6.06 | 0.070 | 0.070 | 35.6 | 2021-02-12 | |

### Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 8-TG-101-01 | 8.18 | 4.06 | 2006-02-24 | 7.89 | 7.76 | 0.022 | 0.013 | >50 | 2026-02-12 | |
| 8-TG-101-02 | 8.18 | 4.06 | 2006-02-24 | 7.75 | 7.46 | 0.037 | 0.029 | >50 | 2026-02-12 | |
| 8-TG-101-03 | 8.18 | 4.06 | 2006-02-24 | 7.79 | 7.54 | 0.035 | 0.025 | >50 | 2026-02-12 | |
| 8-TG-101-04 | 8.18 | 4.06 | 2006-02-24 | 7.76 | 7.51 | 0.030 | 0.025 | >50 | 2026-02-12 | |
| 8-TG-101-05 | 8.18 | 4.06 | 2006-02-24 | 7.95 | 7.73 | 0.025 | 0.022 | >50 | 2026-02-12 | |
| 8-TG-101-07 | 8.18 | 4.06 | — | — | 7.39 | — | — | — | 2026-02-12 | |

### Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-101-01 | 6.02 | 3.18 | 2006-02-24 | 5.51 | 5.02 | 0.049 | 0.049 | 37.4 | 2026-02-12 | |
| 4-LA-101-02 | 6.02 | 3.18 | 2006-02-24 | 5.76 | 5.45 | 0.028 | 0.031 | >50 | 2026-02-12 | |
| 4-LA-101-03 | 6.02 | 3.18 | 2006-02-24 | 5.76 | 5.45 | 0.026 | 0.031 | >50 | 2026-02-12 | |
| 4-LA-101-04 | 6.02 | 3.18 | 2006-02-24 | 5.78 | 5.48 | 0.030 | 0.030 | >50 | 2026-02-12 | |
| 4-LA-101-05 | 6.02 | 3.18 | 2006-02-24 | 5.81 | 5.54 | 0.024 | 0.027 | >50 | 2026-02-12 | |
| 4-LA-101-06 | 6.02 | 3.18 | 2006-02-24 | 5.64 | 5.30 | 0.038 | 0.034 | >50 | 2026-02-12 | |
| 4-LA-101-07 | 6.02 | 3.18 | 2006-02-24 | 5.57 | 4.87 | 0.056 | 0.070 | 24.1 | 2026-02-12 | |
| 4-LA-101-09 | 6.02 | 3.18 | — | — | 5.00 | — | — | — | 2026-02-12 | |

### Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-102-01 | 6.02 | 3.18 | 2006-02-24 | 5.63 | 5.25 | 0.039 | 0.038 | >50 | 2026-02-12 | |
| 4-LA-102-02 | 6.02 | 3.18 | 2006-02-24 | 5.67 | 5.36 | 0.036 | 0.031 | >50 | 2026-02-12 | |
| 4-LA-102-03 | 6.02 | 3.18 | 2006-02-24 | 5.55 | 5.28 | 0.040 | 0.027 | >50 | 2026-02-12 | |
| 4-LA-102-04 | 6.02 | 3.18 | 2006-02-24 | 5.34 | 4.90 | 0.055 | 0.044 | 31.5 | 2026-02-12 | |
| 4-LA-102-05 | 6.02 | 3.18 | 2006-02-24 | 5.62 | 5.34 | 0.033 | 0.028 | >50 | 2026-02-12 | |
| 4-LA-102-06 | 6.02 | 3.18 | 2006-02-24 | 5.83 | 5.60 | 0.020 | 0.023 | >50 | 2026-02-12 | |
| 4-LA-102-07 | 6.02 | 3.18 | 2006-02-24 | 5.63 | 5.40 | 0.030 | 0.023 | >50 | 2026-02-12 | |
| 4-LA-102-08 | 6.02 | 3.18 | — | — | 5.70 | — | — | — | 2026-02-12 | |

### Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-LA-103-01 | 7.11 | 3.56 | 2006-02-24 | 6.75 | 6.44 | 0.029 | 0.031 | >50 | 2026-02-12 | |
| 6-LA-103-02 | 7.11 | 3.56 | 2006-02-24 | 6.87 | 6.53 | 0.033 | 0.034 | >50 | 2026-02-12 | |
| 6-LA-103-03 | 7.11 | 3.56 | 2006-02-24 | 6.65 | 6.28 | 0.040 | 0.037 | >50 | 2026-02-12 | |
| 6-LA-103-04 | 7.11 | 3.56 | 2006-02-24 | 6.52 | 6.15 | 0.051 | 0.037 | >50 | 2026-02-12 | |
| 6-LA-103-05 | 7.11 | 3.56 | 2006-02-24 | 6.49 | 5.98 | 0.061 | 0.051 | 39.9 | 2026-02-12 | |
| 6-LA-103-06 | 7.11 | 3.56 | 2006-02-24 | 6.42 | 5.82 | 0.067 | 0.060 | 33.6 | 2026-02-12 | |
| 6-LA-103-07 | 7.11 | 3.56 | 2006-02-24 | 6.76 | 6.12 | 0.053 | 0.064 | 39.9 | 2026-02-12 | |
| 6-LA-103-08 | 7.11 | 3.56 | 2006-02-24 | 6.79 | 6.49 | 0.028 | 0.030 | >50 | 2026-02-12 | |
| 6-LA-103-10 | 7.11 | 3.56 | — | — | 6.30 | — | — | — | 2026-02-12 | |

### Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-101-01 | 8.56 | 3.81 | 2014-09-19 | 8.07 | 7.83 | 0.035 | 0.098 | 40.9 | 2019-02-12 | |
| 4-RA-101-02 | 8.56 | 3.81 | 2014-09-19 | 8.02 | 7.94 | 0.030 | 0.052 | >50 | 2019-02-12 | |
| 4-RA-101-03 | 8.56 | 3.81 | 2014-09-19 | 7.88 | 7.65 | 0.048 | 0.066 | >50 | 2019-02-12 | |
| 4-RA-101-04 | 8.56 | 3.81 | 2014-09-19 | 8.18 | 8.00 | 0.032 | 0.042 | >50 | 2019-02-12 | |
| 4-RA-101-05 | 8.56 | 3.81 | 2014-09-19 | 7.98 | 7.82 | 0.039 | 0.068 | >50 | 2019-02-12 | |
| 4-RA-101-06 | 8.56 | 3.81 | 2014-09-19 | 7.91 | 7.87 | 0.029 | 0.029 | >50 | 2019-02-12 | |

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-101-07 | 8.56 | 3.81 | 2014-09-19 | 8.22 | 8.14 | 0.023 | 0.032 | >50 | 2019-02-12 | |
| 4-RA-101-08 | 8.56 | 3.81 | 2014-09-19 | 7.71 | 7.31 | 0.069 | 0.142 | 24.6 | 2019-02-12 | |
| 4-RA-101-09 | 8.56 | 3.81 | 2014-09-19 | 8.27 | 8.13 | 0.022 | 0.038 | >50 | 2019-02-12 | |
| 4-RA-101-10 | 8.56 | 3.81 | 2014-09-19 | 8.21 | 8.02 | 0.033 | 0.052 | >50 | 2019-02-12 | |
| 4-RA-101-11 | 8.56 | 3.81 | 2014-09-19 | 7.88 | 7.60 | 0.054 | 0.088 | 42.9 | 2019-02-12 | |

**Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-102-01 | 8.56 | 3.81 | 2014-09-19 | 8.25 | 8.02 | 0.026 | 0.062 | >50 | 2019-02-12 | |
| 4-RA-102-02 | 8.56 | 3.81 | 2014-09-19 | 7.89 | 7.64 | 0.047 | 0.070 | >50 | 2019-02-12 | |
| 4-RA-102-03 | 8.56 | 3.81 | 2014-09-19 | 7.95 | 7.55 | 0.051 | 0.098 | 38.0 | 2019-02-12 | |
| 4-RA-102-04 | 8.56 | 3.81 | 2014-09-19 | 7.93 | 7.72 | 0.041 | 0.048 | >50 | 2019-02-12 | |
| 4-RA-102-05 | 8.56 | 3.81 | 2014-09-19 | 7.84 | 7.67 | 0.044 | 0.080 | 48.1 | 2019-02-12 | |
| 4-RA-102-06 | 8.56 | 3.81 | 2014-09-19 | 7.95 | 7.73 | 0.039 | 0.056 | >50 | 2019-02-12 | |
| 4-RA-102-07 | 8.56 | 3.81 | 2014-09-19 | 7.98 | 7.78 | 0.040 | 0.064 | >50 | 2019-02-12 | |
| 4-RA-102-08 | 8.56 | 3.81 | 2014-09-19 | 7.58 | 7.35 | 0.057 | 0.106 | 33.3 | 2019-02-12 | |
| 4-RA-102-09 | 8.56 | 3.81 | 2014-09-19 | 7.84 | 7.54 | 0.051 | 0.088 | 42.2 | 2019-02-12 | |
| 4-RA-102-11 | 8.56 | 3.81 | 2014-09-19 | 8.12 | 7.96 | 0.046 | 0.046 | >50 | 2019-02-12 | |

**Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-RA-103-01 | 10.97 | 4.57 | 2014-09-19 | 10.62 | 10.29 | 0.036 | 0.036 | >50 | 2019-02-12 | |
| 6-RA-103-02 | 10.97 | 4.57 | 2014-09-19 | 10.06 | 9.69 | 0.070 | 0.108 | 47.2 | 2019-02-12 | |
| 6-RA-103-03 | 10.97 | 4.57 | 2014-09-19 | 10.39 | 10.06 | 0.047 | 0.072 | >50 | 2019-02-12 | |
| 6-RA-103-04 | 10.97 | 4.57 | 2014-09-19 | 9.79 | 9.31 | 0.084 | 0.132 | 35.8 | 2019-02-12 | |
| 6-RA-103-05 | 10.97 | 4.57 | 2014-09-19 | 9.81 | 9.48 | 0.074 | 0.124 | 39.5 | 2019-02-12 | |
| 6-RA-103-06 | 10.97 | 4.57 | 2014-09-19 | 10.43 | 10.32 | 0.035 | 0.048 | >50 | 2019-02-12 | |
| 6-RA-103-07 | 10.97 | 4.57 | 2014-09-19 | 10.37 | 10.20 | 0.040 | 0.070 | >50 | 2019-02-12 | |
| 6-RA-103-08 | 10.97 | 4.57 | 2014-09-19 | 10.49 | 10.33 | 0.032 | 0.022 | >50 | 2019-02-12 | |
| 6-RA-103-09 | 10.97 | 4.57 | 2014-09-19 | 10.39 | 10.33 | 0.034 | 0.046 | >50 | 2019-02-12 | |
| 6-RA-103-10 | 10.97 | 4.57 | 2014-09-19 | 9.93 | 9.49 | 0.077 | 0.112 | 43.8 | 2019-02-12 | |
| 6-RA-103-11 | 10.97 | 4.57 | 2014-09-19 | 10.10 | 9.77 | 0.236 | 0.236 | 22.0 | 2019-02-12 | |
| 6-RA-103-13 | 10.97 | 4.57 | 2014-09-19 | 10.38 | 10.15 | 0.060 | 0.060 | >50 | 2019-02-12 | |

**Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-RV-101-01 | 7.62 | 3.56 | 2013-09-09 | 6.49 | 6.34 | 0.063 | 0.081 | 34.2 | 2019-02-12 | |
| 3-RV-101-02 | 7.62 | 3.56 | 2013-09-09 | 6.44 | 6.31 | 0.063 | 0.052 | 43.5 | 2019-02-12 | |
| 3-RV-101-03 | 7.62 | 3.56 | 2013-09-09 | 5.80 | 5.63 | 0.095 | 0.098 | 21.2 | 2019-02-12 | |
| 3-RV-101-04 | 7.62 | 3.56 | 2013-09-09 | 6.13 | 5.88 | 0.086 | 0.102 | 22.8 | 2019-02-12 | |
| 3-RV-101-05 | 7.62 | 3.56 | 2013-09-09 | 6.48 | 6.36 | 0.066 | 0.050 | 42.6 | 2019-02-12 | |
| 3-RV-101-06 | 7.62 | 3.56 | 2013-09-09 | 6.11 | 5.96 | 0.082 | 0.087 | 27.7 | 2019-02-12 | |
| 3-RV-101-07 | 7.62 | 3.56 | 2013-09-09 | 5.73 | 5.54 | 0.122 | 0.122 | 16.2 | 2019-02-12 | |
| 3-RV-101-08 | 7.62 | 3.56 | 2013-09-09 | 6.36 | 6.31 | 0.074 | 0.074 | 37.2 | 2019-02-12 | |

**Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RV-102-01 | 5.54 | 2.79 | 2013-09-09 | 4.49 | 4.35 | 0.059 | 0.037 | 26.4 | 2019-02-12 | |
| 2-RV-102-02 | 5.54 | 2.79 | 2013-09-09 | 4.29 | 4.01 | 0.079 | 0.109 | 11.2 | 2019-02-12 | |
| 2-RV-102-03 | 5.54 | 2.79 | 2013-09-09 | 4.68 | 4.64 | 0.042 | 0.037 | 43.9 | 2019-02-12 | |

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RV-102-04 | 5.54 | 2.79 | 2013-09-09 | 4.81 | 4.73 | 0.043 | 0.020 | 45.5 | 2019-02-12 | |
| 2-RV-102-05 | 5.54 | 2.79 | 2013-09-09 | 4.96 | 4.97 | 0.013 | 0.013 | >50 | 2019-02-12 | |
| 2-RV-102-06 | 5.54 | 2.79 | 2013-09-09 | 4.55 | 4.27 | 0.102 | 0.102 | 14.6 | 2019-02-12 | |

**Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-RF-101-01 | 5.49 | 2.41 | 2013-12-09 | 5.37 | 5.31 | 0.009 | 0.017 | >50 | 2026-02-12 | |
| 3-RF-101-02 | 5.49 | 2.41 | 2013-12-09 | 5.40 | 5.34 | 0.006 | 0.003 | >50 | 2026-02-12 | |
| 3-RF-101-03 | 5.49 | 2.41 | 2013-12-09 | 5.37 | 5.33 | 0.006 | 0.012 | >50 | 2026-02-12 | |
| 3-RF-101-04 | 5.49 | 2.41 | 2013-12-09 | 5.41 | 5.35 | 0.006 | -0.002 | >50 | 2026-02-12 | |
| 3-RF-101-05 | 5.49 | 2.41 | 2013-12-09 | 5.32 | 5.33 | 0.006 | 0.007 | >50 | 2026-02-12 | |
| 3-RF-101-06 | 5.49 | 2.41 | 2013-12-09 | 5.17 | 5.23 | 0.015 | 0.017 | >50 | 2026-02-12 | |
| 3-RF-101-07 | 5.49 | 2.41 | 2013-12-09 | 5.48 | 5.37 | 0.003 | -0.002 | >50 | 2026-02-12 | |
| 3-RF-101-09 | 5.49 | 2.41 | 2013-12-09 | 5.26 | 5.19 | -0.029 | -0.029 | >50 | 2026-02-12 | |

**Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RF-102-01 | 3.91 | 2.16 | 2013-12-09 | 3.60 | 3.76 | 0.008 | 0.009 | >50 | 2026-02-12 | |
| 2-RF-102-02 | 3.91 | 2.16 | 2013-12-09 | 3.70 | 3.64 | 0.012 | 0.024 | >50 | 2026-02-12 | |
| 2-RF-102-03 | 3.91 | 2.16 | 2013-12-09 | 3.79 | 3.75 | 0.009 | 0.005 | >50 | 2026-02-12 | |
| 2-RF-102-04 | 3.91 | 2.16 | 2013-12-09 | 3.82 | 3.38 | 0.025 | 0.043 | 28.3 | 2026-02-12 | |
| 2-RF-102-05 | 3.91 | 2.16 | 2013-12-09 | 3.67 | 3.49 | 0.082 | 0.082 | 16.1 | 2024-03-06 | |
| 2-RF-102-06 | 3.91 | 2.16 | 2013-12-09 | 3.67 | 3.47 | 0.099 | 0.099 | 13.2 | 2022-09-26 | |
| 2-RF-102-07 | 3.91 | 2.16 | 2013-12-09 | 3.72 | 3.57 | 0.020 | 0.028 | >50 | 2026-02-12 | |

**Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-104-01 | 6.02 | 3.18 | 2006-02-24 | 5.75 | 5.31 | 0.038 | 0.044 | 48.2 | 2026-02-12 | |
| 4-LA-104-02 | 6.02 | 3.18 | 2006-02-24 | 5.56 | 5.00 | 0.052 | 0.056 | 32.4 | 2026-02-12 | |
| 4-LA-104-03 | 6.02 | 3.18 | 2006-02-24 | 5.74 | 5.42 | 0.027 | 0.032 | >50 | 2026-02-12 | |
| 4-LA-104-04 | 6.02 | 3.18 | 2006-02-24 | 5.86 | 5.63 | 0.022 | 0.023 | >50 | 2026-02-12 | |
| 4-LA-104-05 | 6.02 | 3.18 | 2006-02-24 | 5.51 | 4.88 | 0.058 | 0.063 | 26.9 | 2026-02-12 | |
| 4-LA-104-06 | 6.02 | 3.18 | 2006-02-24 | 5.83 | 5.55 | 0.028 | 0.028 | >50 | 2026-02-12 | |
| 4-LA-104-07 | 6.02 | 3.18 | 2006-02-24 | 5.62 | 5.28 | 0.037 | 0.034 | >50 | 2026-02-12 | |
| 4-LA-104-08 | 6.02 | 3.18 | 2006-02-24 | 5.50 | 5.11 | 0.042 | 0.039 | 46.3 | 2026-02-12 | |
| 4-LA-104-10 | 6.02 | 3.18 | — | — | 4.84 | — | — | — | 2026-02-12 | |

**Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-104-01 | 8.56 | 3.81 | 2014-09-19 | 7.84 | 7.71 | 0.040 | 0.048 | >50 | 2019-02-12 | |
| 4-RA-104-02 | 8.56 | 3.81 | 2014-09-19 | 8.09 | 7.85 | 0.035 | 0.058 | >50 | 2019-02-12 | |
| 4-RA-104-03 | 8.56 | 3.81 | 2014-09-19 | 8.46 | 8.28 | 0.057 | 0.056 | >50 | 2019-02-12 | |
| 4-RA-104-04 | 8.56 | 3.81 | 2014-09-19 | 8.49 | 8.18 | 0.063 | 0.084 | >50 | 2019-02-12 | |
| 4-RA-104-05 | 8.56 | 3.81 | 2014-09-19 | 8.37 | 8.16 | 0.050 | 0.082 | >50 | 2019-02-12 | |
| 4-RA-104-06 | 8.56 | 3.81 | 2014-09-19 | 7.91 | 7.61 | 0.049 | 0.074 | >50 | 2019-02-12 | |
| 4-RA-104-07 | 8.56 | 3.81 | 2014-09-19 | 7.61 | 7.32 | 0.064 | 0.092 | 38.0 | 2019-02-12 | |
| 4-RA-104-08 | 8.56 | 3.81 | 2014-09-19 | 8.00 | 7.81 | 0.074 | 0.074 | >50 | 2019-02-12 | |

**Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-ST-101-01 | 3.91 | 2.29 | 2006-02-24 | 3.64 | 3.61 | 0.016 | 0.003 | >50 | 2026-02-12 | |
| 2-ST-101-02 | 3.91 | 2.29 | 2006-02-24 | 3.67 | 3.55 | 0.018 | 0.012 | >50 | 2026-02-12 | |
| 2-ST-101-03 | 3.91 | 2.29 | 2006-02-24 | 3.82 | 3.55 | 0.020 | 0.027 | 46.5 | 2026-02-12 | |
| 2-ST-101-04 | 3.91 | 2.29 | 2006-02-24 | 3.77 | 3.68 | 0.007 | 0.009 | >50 | 2026-02-12 | |
| 2-ST-101-05 | 3.91 | 2.29 | 2006-02-24 | 3.72 | 3.50 | 0.021 | 0.022 | >50 | 2026-02-12 | |
| 2-ST-101-06 | 3.91 | 2.29 | — | — | 3.63 | — | — | — | 2026-02-12 | |

**Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-CW-101-01 | 5.49 | 2.79 | 2006-02-24 | 5.13 | 5.03 | 0.020 | 0.010 | >50 | 2026-02-12 | |
| 3-CW-101-02 | 5.49 | 2.79 | 2006-02-24 | 5.20 | 5.01 | 0.020 | 0.019 | >50 | 2026-02-12 | |
| 3-CW-101-03 | 5.49 | 2.79 | 2006-02-24 | 5.08 | 4.93 | 0.030 | 0.015 | >50 | 2026-02-12 | |
| 3-CW-101-04 | 5.49 | 2.79 | 2006-02-24 | 5.38 | 5.26 | 0.010 | 0.012 | >50 | 2026-02-12 | |
| 3-CW-101-05 | 5.49 | 2.79 | 2006-02-24 | 5.33 | 5.35 | 0.007 | -0.002 | >50 | 2026-02-12 | |
| 3-CW-101-06 | 5.49 | 2.79 | — | — | 5.08 | — | — | — | 2026-02-12 | |

**Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-CW-102-01 | 5.49 | 2.79 | 2006-02-24 | 5.20 | 5.06 | 0.022 | 0.014 | >50 | 2026-02-12 | |
| 3-CW-102-02 | 5.49 | 2.79 | 2006-02-24 | 5.02 | 4.37 | 0.057 | 0.065 | 24.2 | 2026-02-12 | |
| 3-CW-102-03 | 5.49 | 2.79 | 2006-02-24 | 5.07 | 4.54 | 0.050 | 0.053 | 32.9 | 2026-02-12 | |
| 3-CW-102-04 | 5.49 | 2.79 | 2006-02-24 | 5.03 | 4.69 | 0.041 | 0.034 | 46.2 | 2026-02-12 | |
| 3-CW-102-05 | 5.49 | 2.79 | — | — | 4.66 | — | — | — | 2026-02-12 | |

| Sign-off | Name | Function | Date | Signature |
|---|---|---|---|---|
| Prepared | Marcus Chen | lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II | 2016-02-12 | _____ |